The Land, Water, and Power a Data Center Uses Are the Wrong Things to Fear

A 785,000-square-foot data center sitting on 217 acres, drawing 1.31 billion kilowatt-hours of electricity a year and 1.8 million gallons of water a day, sounds like an environmental monstrosity when these figures are printed in the headlines. But most mainstream media are not showing you the big picture; raw resource consumption, isolated in a headline, provides insufficient backing to base decision making off of. The question we should be asking is: what do we get in return, and how does that compare to the alternatives competing for the same land, water, and power?

Highlight the comparison, and the picture changes. Your average 18-hole country club occupies roughly the same 200-acre footprint as a data center facility, but its few dozen jobs supplies doesn't compete with the hundreds or thousands of downstream jobs a data center creates. A modest 440-acre alfalfa farm in the Southwest consumes the same 1.8 million gallons a day that a data center would, yet returns well under a million dollars a year in crop value, a meer rounding error compared to the annual revenue generated by data centers and their services. A large bitcoin mining operation can draw the same 1.31 billion kilowatt-hours a year off the grid, while contributing essentially nothing to the economy based on standard metrics[1]. None of these comparisons make a data center look wasteful, instead, they highlight that we may be caring about the wrong things.

Instead of looking at all the negatives (costs) associated with our example data center, we reframe and analyze the benefits (utilities) of such an entity. A golf course, a farm, and a mining rig all convert resources into a *bounded, terminal* output, whereas a data center converts the same resources into *general-purpose computing capacity*. Other entities, across many sectors like finance, healthcare, logistics, retail, and research benefit from this utility, and the data center creates a trickle down effect through these sectors. This is the same economic logic that made investments in electrification and the interstate highway system pay off far beyond their direct footprint: the marginal cost is paid on-site, but the marginal benefit compounds downstream, in industries the facility itself never touches.

None of this means data centers should get away easily. The negative externalities of the infrastructure needed to power these data centers are concentrated locally; on local grids, waterways, land, and environment. Their benefits, however, are diffuse and often seen elsewhere. This disconnect between who sees the benefits versus the costs is a legitimate basis for considering whether or not to approve the project and communities are right to ask the hard questions. However, the answer to those questions shouldn't be to shoot down the idea just because of some large figures on a utility bill. Numbers alone can be scary, but when we put them into perspective, the benefits a data center can bring to a region bring to light the true situation we must thoughtfully consider.

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[1] Standard metrics like local employment and GDP contribution per kWh